SUBJCTIVES OF CURRENT PAPER

BT302 - IMMUNOLOGY

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Types of active immunization vaccines:

Active immunization is the induction of immunity after exposure to an antigen, requiring antigenic exposure. It occurs naturally or artificially, leading to antibody development and memory response. The immune system generates antibodies slowly, providing long-term immunity without pre-made antibodies in the host's serum.

Types: or Examples: Active immunization vaccines include live attenuated vaccines (e.g., MMR, varicella), inactivated vaccines (e.g., polio, hepatitis A), subunit vaccines (e.g., hepatitis B, HPV), conjugate vaccines (e.g., Hib), toxoid vaccines (e.g., tetanus, diphtheria), and mRNA vaccines (e.g., COVID-19).

What is Opsonization?

Opsonization is the enhancement of the phagocytic process where pathogens are marked for ingestion by phagocytes. Opsonins, such as immunoglobulins (IgG, IgM) and complement proteins, bind to the bacterial surface, facilitating neutrophil or macrophage phagocytosis through opsonin receptors like Fc and complement receptors.

The structure of TCR

The T-cell receptor (TCR) is a heterodimer surface receptor composed of two equal-length chains (α and β). Each chain has a constant (C) and variable (V) region, with hypervariable regions for specificity. TCRs have a short cytoplasmic tail, transmembrane region, and disulfide bridge.

Characteristics of immunoglobulin:

Immunoglobulins (Ig) are antibodies produced by plasma cells, composed of two heavy and two light chains forming a Y-shaped molecule. They have variable regions for antigen binding, and constant regions for effector functions. Classes include IgG, IgA, IgM, IgE, and IgD, each with specific roles and functions.

Parts of secondary lymphoid system?

The secondary lymphoid system includes structures where lymphocytes become activated and initiate adaptive immune responses. Key parts of this system are:

1. Lymph Nodes: A=Filter lymph fluid, b=Sites for antigen presentation and lymphocyte activation.

2. Spleen: A= Filters blood, B= Red pulp: Removes old red blood cells, c= White pulp: Immune responses to blood-borne antigens.

3. Tonsils: A= Located at the throat and nasal passages. B=First line of defense against inhaled or ingested pathogens.

4. Appendix: A=Contains lymphoid tissue. B= Plays a role in gut immunity.

Complement System:

The complement system is a group of proteins that enhance the immune response. These proteins can directly attack pathogens, promote phagocytosis, and contribute to the inflammatory response. Among complement proteins, some proteins have the ability to bind with immunoglobulin molecules while some are involved in binding on the membrane component of various immune cells.

Endogenous & exogenous antigen difference?

Endogenous and exogenous antigens are two main types of antigens that the immune system recognizes and responds to:

Endogenous Antigens:

- Origin: Generated within the body's cells.
- Produced within the cell, presented by MHC Class I, and targeted by CD8+ T cells.
- Examples: Viral proteins produced inside infected cells, tumor antigens from cancerous cells, and normal cellular proteins presented during routine maintenance.

Exogenous Antigens:

- Origin: Enter the body from the external environment.
- presented by MHC Class II, and targeted by CD4+ T cells.
- Examples: Bacteria, fungi, parasites, pollen, and toxins.

Activation of B-cell receptor?

B-cells are activated when there is high concentration of antigen for example in case of bacteria the lipopolysacchrides antigens exist in greater quantity. B-cell activation occur after binding of antigen to a surface molecule which is known as B-cell receptor. Afterward there is bypassing of the early part of the biochemical pathway mediated by specific antigen receptor. Cross linking of immunoglobulin receptor occur due to repeating determinants on antigenic structure. As a result the transformation of B-cells occur into plasma cell for immunoglobulin secretion.

Active / passive immunization?

1= Active immunization: It is the induction of immunity after exposure of host to antigen. In this case, antigenic exposure is mandatory. Active immunization can occur naturally by exposing to microbe or other antigen when there is no prior exposure before the entry of antigen. In this form of immunization, immune system develops antibodies against the microbe after its exposure. This process of antibodies generation is considered slow.

2= Passive immunization: It is the induction of active humoral immunity. In this kind of immunization, antibodies are mandatory. Passive immunization can occur naturally by transfer of maternal antibodies to fetus through placenta. However, the artificial induction can be done by injecting gamma globulins from other individuals or animals to host.

Adaptive immune system?

- Response is antigen-dependent
- There is a lag time between exposure and maximal response
- Antigen-specific
- Exposure results in immunologic memory
- Specific
- Second line of Defense
- Needs to be activated

What are cytokines?

• Diverse group of non-antibody proteins • Act as mediators between cells • Initially these were considered as products of immune cells • Also considered to act as mediator for immune cells • Also produced from non-immune cells & mediators for them. • Used as biological response

modifier for treatment of various diseases • Cytokine is a general term but more specific term used according to their cell of origin.

What are monoclonal antibodies?

The antibodies which are produced from a single antibody producing B-cell are called as monoclonal antibodies. These antibodies also able to bind with single and unique antigen binding sit (epitope). Monoclonal antibodies mainly consist of single subtype of IgG for example IgG1, IgG2 & IgG3.

Control of Intracellular Pathogens / How immune system responds against intracellular pathogens?

The secretory immune molecules are ineffective against intracellular pathogens like viruses and intracellular bacteria. The cell mediated immune response is the primary defense against intracellular pathogens. The T-Lymphocytes play vital role in cell mediated immunity. Cell mediated immune response varies according to the residing site of the pathogen:

1) Cytosolic site (The exogenous pathogens are the pathogens which used to reside in the cytosol of infected cell)

2) Vesicular site (The endogenous pathogens are those pathogens which used to reside in the vesicle like phagosomes of infected cell)

According to time of rejection how many types of graft rejection:

According to time of rejection: graft rejection can be of following types with distinct immune mechanisms

1) Hyper acute rejection

1) Accelerated rejection

1) Acute Rejection

1) Chronic Rejection

Cytokine in adaptive immune system:

Followings are the cytokines which act primarily for Adaptive (Specific) immune system

1) IL-2 It is produced by T-helper (Th) cells but Cytotoxic T-cells (Tc) also produced it in lesser extent. It stimulate B-cell division. It can also activate NK cells and monocytes. Moreover, IL-2 acts on Tcells in autocrine manner.

2) IL-4 IL-4 is produced by macrophages and Th2 cells. It promotes the development of Th2 from naïve T-helper (Th) cells. The Differentiated Th2 cells result in the production of antibodies. IL-4 also involves in the class switching of antibody into IgE

3) IL-5 IL-5 is produced by Th2 cells. It promotes the development of B-cells and eosinophils. It also activates mature eosinophils.

4) Tumor Growth Factor (TGF)-β TGF- β is produced by T cells and many other cells types. It acts as inhibitory cytokine which inhibits the proliferation of T-cells and activation of macrophages. It also acts on neutrophils and endothelial cells to block the effects of pro-inflammatory cytokines.

Natural active immunization;

Natural immunization is acquired immunization which is achieved by natural exposure to antigen or acquiring infection by an infectious agent or microorganism e.g measles virus. This is also termed as natural active immunization. This form of natural exposure to antigen stimulate the humoral immunity as a result specific antibodies are generated against exposed microbe. Moreover, there is also activation of cell mediated immunity for intracellular pathogen.

Artificial Active Immunization:

Artificial immunization is the kind of immunization which is acquired by artificial exposure to antigen or microorganism which is administrated to body in less virulent or attenuated form like heat killed forms of microbes. This artificial exposure with antigen stimulate the humoral immunity in the form antibodies generation. Moreover, artificial activation of cell mediated immunity for intracellular pathogen for example BCG for Mycobacterium tuberculosis. Artificial immunization: acquired immunization by artificial exposure. This artificial exposure to antigen or microbe which is by administration of less virulent (attenuated), heat killed microbes to organism. Artificial exposure to antigen stimulate the humoral immunity for antibodies.

Importance of immune system?

The immune system plays a vital role in protecting the body against infections, diseases, and foreign substances. Here are some key importance of the immune system:

1. Protection against infections: The immune system defends against bacteria, viruses, fungi, and other pathogens that can cause infections.

2. Cancer prevention: The immune system can recognize and eliminate cancer cells, preventing tumor growth and development.

3. Tissue repair: The immune system helps repair damaged tissues and promotes healing.

4. Detoxification: The immune system removes toxins and foreign substances from the body.

5. Allergy prevention: The immune system regulates allergic responses, preventing overreactions to harmless substances.

6. Maintenance of homeostasis: The immune system helps maintain the body's internal balance and stability.

Signs of inflammation?

Signs of inflammation: pain, heat, redness, swelling & loss of tissue functions.

Inflammation can be beneficial for host in order to remold the damaged tissue • Undesirable effects: in the form of signs of inflammation • Dolor (Pain) • Calor (Heat) • Rubor (Redness) • Tumor (Swelling) • Functio laesa (Loss of function).

Define myelin stem cell:

A myelin stem cell is a type of progenitor cell capable of differentiating into myelin-producing cells, such as oligodendrocytes in the central nervous system and Schwann cells in the peripheral nervous system. These cells play a crucial role in repairing and regenerating damaged myelin sheaths.

Define antigen, antibodies and heptane ?

1= Antigen A substance that reacts with the products of specific immune response.

2= Immunogen A substance that induces specific immune response.

3= Hapten A substance that is nonimmunogenic but can react with products of specific immune response.

Explain autoimmunity.

Autoimmunity refers to the breakdown of self-tolerance, causing immune responses against self-antigens. This leads to harmful reactions involving antibodies and T-cells, damaging host tissues. Genetic predispositions (e.g., immunoglobulin, TCR, MHC genes) and environmental factors (e.g., drugs, infections) contribute. Mechanisms include sequestered antigens, escape of autoreactive clones, lack of regulatory T-cells, and cross-reactive antigens.

Factors that affect antibody and antigen test:

Factors affecting antigen and antibody tests include:

1. Affinity: The strength of binding between a single antigenic determinant and a single antibody site. Higher affinity results in more stable interactions. 2. Avidity: The overall strength of binding between multivalent antigens and antibodies, influenced by the valency of both. Greater avidity leads to more stable and easily detectable reactions. 3. Antigen & Antibody Ratio: The ratio determines the nature of the complex, with equivalence facilitating lattice formation and larger complexes being more detectable. 4. Physical Form: The physical form of the antigen affects the reaction type, with particulate antigens causing agglutination and soluble antigens causing precipitation.

What is the use of monoclonal antibodies:

Monoclonal antibodies are used for diagnosing and treating diseases. They target specific antigens, making them effective in cancer therapy, autoimmune diseases, infectious diseases, and as diagnostic tools in imaging and laboratory tests, offering precise and targeted medical interventions.

Define phagocytosis. Explain the process of phagocytosis:

Phagocytosis is a vital cellular process where specialized cells called phagocytes engulf and digest solid particles like bacteria and cellular debris.

The process involves several stages:

1. Chemotaxis: Phagocytes move towards infection sites due to chemical signals. 2. Attachment: Phagocytes adhere to foreign particles using specific receptors. 3. Engulfment: Phagocytes extend their membrane to surround the particle, forming a phagosome. 4. Phagosome Formation: The phagosome matures and fuses with lysosomes. 5. Digestion: Lysosomal enzymes break down the phagosome's contents. 6. Exocytosis: Indigestible material is expelled from the cell.

Which theories explain the autoimmunity:

Autoimmunity theories include sequestered antigens, where certain self-antigens are confined to specialized organs and may trigger autoimmune responses if released due to injury. Another theory is the escape of autoreactive cellular clones, involving the loss of central tolerance allowing autoreactive T-cells and B-cells to escape deletion. The lack of regulatory T-cells, leading to insufficient suppression of immune responses, and cross-reactive antigens, where pathogen antigens resemble self-antigens and trigger autoimmune responses, are also key theories explaining autoimmunity.

State the principals of complement reaction test:

The principles of the complement reaction test include antigen-antibody interaction, where the presence of specific antibodies binds to the antigen. This complex activates the complement system, a group of proteins that enhance immune responses. The test typically measures the extent of hemolysis or other reactions caused by complement activation. It's used to detect the presence of antibodies or antigens, diagnosing infections and autoimmune diseases by observing complement activity.

What is cytokines network and how it's related to lymphocytes, macrophage and other immune response components?

The cytokine network is a complex series of interrelated connections among cytokines, where cytokines secreted from one type of cell affect other cells and organs. Within this network, cytokines can have synergistic or antagonistic effects on each other. This communication involves lymphocytes, macrophages, the hypothalamus, kidneys, liver, and other tissues, coordinating immune responses. Cytokines regulate immune responses at the recognition, activation, and effector phases, acting as positive or negative regulators based on the presence of other cytokines and receptor expression on effector cells, balancing antigen-driven activation and regulatory influences.

Differentiate between T-dependent and T-independent antigens.

T-cell dependent antigens: These are antigens which need the help of T-cells for mounting an immune response otherwise these cannot. T-dependent requires the help of T-cells for activation of B-cells. Such kinds of antigens are protein in nature. Moreover, these antigens carry variety of antigenic determinants (epitopes) with few copies.

T-Cell independent antigens: These antigens are those antigens which can directly activate Bcells without the help of T-cells. Such kinds of antigens are non-protein and usually are polysaccharides in nature. These antigens carry same kind of epitopes in polymeric form (Figure 3.3). Moreover, such kinds of antigens are most resistant to degradation.

Cell of cellular barrier:

Cells of the cellular barrier are key components of the immune system's first line of defense.

1. Neutrophils: Rapidly respond to infections, phagocytose bacteria, release antimicrobials. 2. Macrophages: Phagocytose pathogens, present antigens, secrete cytokines. 3. Dendritic Cells: Capture and present antigens to T cells. 4. NK Cells: Destroy infected/cancerous cells by inducing apoptosis. 5. Eosinophils: Combat parasitic infections, participate in allergic reactions.

Write mechanism of activation of B cell with the independent antigen:

The activation of B cells with T-independent antigens involves direct binding of the antigen to the B cell receptor (BCR), cross-linking multiple BCRs. This triggers intracellular signaling, leading to B cell proliferation and differentiation into plasma cells, which produce antibodies specific to the antigen.

What is direct and indirect ELISA?

Direct ELISA • Unknown Ab can be determined • Known Ag is coated on solid phase (Glass surface) • Serum containing unknown Ab is added • Another ab which is precoated with enzyme e.g horseradish peroxidase • Substrate:H2O2

Indirect (Sandwich) ELISA • Unknown Ag can be determined • Known Ab is coated on solid phase (Glass surface) • Serum containing unknown Ag is added • Another Ab which is precoated with enzyme e.g horseradish peroxidase • Substrate: H2O2

Co-stimulatory of B-cell antibodies?

CD40: Interaction with CD40 ligand (CD40L) on T cells provides essential costimulatory signals for B cell activation.

B7 Molecules (CD80/CD86): Interaction with CD28 on T cells enhances B cell activation and antibody production. ICOS (Inducible T-cell Co-stimulator): Interaction with its ligand on B cells promotes T cell-dependent B cell responses.

BAFF (B-cell Activating Factor): Stimulates B cell survival, proliferation, and antibody production.

APRIL (A Proliferation-Inducing Ligand): Enhances B cell survival and immunoglobulin class-switching.

How immune system responses to tumor cell:

The immune system responds to tumor cells through various mechanisms. Tumor-infiltrating lymphocytes (TILs) and other immune cells like macrophages and dendritic cells infiltrate tumors. Elevated levels of cytokines, chemokines, and checkpoint proteins (PD-1, PD-L1, CTLA-4) are observed. Tumor-specific antibodies, both circulating and autoantibodies, target tumor antigens. Immunotherapy can cause immune-related adverse events (irAEs) and lead to tumor regression. Techniques like immunohistochemistry (IHC) and flow cytometry identify immune cells in tumors. Molecular evidence includes high neoantigen load and upregulation of immune-related genes, indicating an active immune response.

Immune evasion mechanism:

Immune evasion mechanisms are strategies used by pathogens and tumor cells to evade the host immune system. These include antigenic variation, where organisms mutate their surface molecules, and inhibition of complement activation by degrading complement proteins. Pathogens may resist phagocytosis through cell surface molecules or toxins that kill phagocytes. Extracellular organisms inactivate humoral immune components, while intracellular organisms bypass cell-mediated immunity. Examples include the antigenic shift and drift in influenza, and resistance mechanisms by bacteria like S. aureus and Pneumococcus to avoid immune detection and destruction.

.Onco fetal antigen?

The tumor antigens which are also expressed by normal cells. These are expressed in high levels by tumor cells as compare to normal cells. These kind of antigens are also called as onco-fetal antigens as these are expressed during early development & lost during adult life. There are following two types of onco-fetal antigens

1) Alpha-fetoproteins (AFP)

2) Carcino-embryonic antigens (CEA)

What is affinity maturation?

The affinity of antibodies depend upon the dosage of antigen, the low dosage of antigen causes increase in the affinity of antibodies which is called as affinity maturation.

Three loci Classes of II MHC?

The class II MHC is encoded by genes for Human MHC which are located at chromosome 6. Structurally, the Class II MHC complex is also composed of three following major **loci DP, DQ, DR**. The Class II MHC is expressed on B-lymphocytes and antigen presenting cells (APC).

What is secondary lymphoid organs.

Secondary lymphoid organs, like lymph nodes, spleen, and mucosal-associated lymphoid tissues (MALT), including gut-associated lymphoid tissue (GALT), trap antigens and facilitate interaction between mature lymphocytes and antigens. They play a crucial role in the adaptive immune response by facilitating immune cell activation and proliferation.

Natural and artificial immunity:

Natural immunity occurs through direct exposure to pathogens, leading to disease and recovery, which creates immunity. Artificial immunity is achieved through medical interventions like vaccines or antibody treatments. Natural immunity develops from infection, while artificial immunity is induced by immunization or therapeutic agents.

On which basis the antibodies are divided into types ? Explain the types and subtypes of the antibodies.

Antibodies are divided into types based on their heavy chain structure and function. The five main types are:

1. IgG: Monomer, provides long-term immunity and crosses the placenta. Subtypes: IgG1, IgG2, IgG3, IgG4.

2. IgA: Usually a dimer, protects mucosal surfaces and is found in secretions. Subtypes: IgA1, IgA2.

3. IgM: Pentamer, first antibody produced in the initial immune response, no subtypes.

4. IgE: Monomer, involved in allergic reactions and defense against parasites, no subtypes.

5. IgD: Monomer, acts as a receptor on naïve B cells, no subtypes.

How parasitic activity is caused by mast cells:

Mast cells play a crucial role in combating parasitic infections through their release of mediators such as histamine, proteases, and cytokines. Upon detecting parasites, mast cells degranulate, releasing these substances to induce inflammation and recruit other immune cells. This response helps to expel parasites by causing increased mucus production, enhancing phagocytosis, and promoting a localized inflammatory reaction. Mast cells also contribute to tissue repair and immune regulation.

What is immunodeficiency:

Immunodeficiency is a condition where the immune system's ability to respond to infections or diseases is impaired. It can be caused by genetic defects, infections, or other factors, leading to increased susceptibility to infections and poor disease resistance.

How antibodies formed:

Antibodies are formed through an immune response where B-cells recognize antigens with specific receptors. These B-cells transform into plasma cells, which then secrete antibodies. The process can be T-independent or T-dependent.

What is organ specific autoimmunity and organ non specific autoimmunity:

Organ-specific autoimmunity involves immune responses targeting specific organ-associated antigens, damaging only the affected organ. Examples include Hashimoto's thyroiditis. Non-organ-specific autoimmunity, or systemic autoimmunity, involves widespread immune responses against non-organ-specific antigens, affecting multiple organs and tissues, such as in systemic lupus erythematosus.

Short difference between TCR and bcr:

T-Cell Receptors (TCR) primarily recognize protein antigens presented by MHC molecules on nucleated cells, categorizing T cells based on MHC recognition. In contrast, B-Cell Receptors (BCR) recognize soluble antigens directly, facilitating antibody production. TCRs rely on MHC presentation, while BCRs reflect the specificity of the antibodies they produce.

What are defects of acute inflammation?

Defects in acute inflammation include insufficient inflammation (e.g., immunodeficiency), excessive inflammation (e.g., chronic inflammation, autoimmune disorders), defective inflammatory mediators (e.g., cytokine dysregulation), impaired phagocytosis, defective resolution of inflammation, and vascular defects (e.g., abnormal permeability).

Role of Cytotoxic T-Cells?

The cytotoxic T-Lymphocytes are also called as CTLs and CD8+ve. These cells are not mature after exiting thymus where these have functional TCR which can recognize antigens but cannot kill the target cells. There is needs of differentiation for fully activation of CTLs. CTL differentiates from Pre-CTLs in response to following two signals

1. The specific antigen in context of Class I MHC

2. The cytokines produced from T-helper 1 (Th1) cells especially IL-2 & INF-γ

VJD Recombination:

V(D)J recombination is a vital process in generating antibody diversity. It involves the joining of various gene segments, such as V and J in light chain genes and V, D, and J in heavy chain genes. Recombination is regulated by Recombination Signal Sequences (RSS), consisting of conserved haptamer and nonamer sequences separated by 2 and 1 turn signals. Enzymes like RAG1 and RAG2 facilitate this recombination process, crucial for the immune system's functionality.

Explain briefly pathogen associated molecular patterns (PAMP).

The antigenic determinant recognized by innate immune system is purely non-specific in nature due to nonspecific attribute of innate immune system. These determinants are present on the

surface of the pathogens in a form of various molecular configurations called as pathogen associated molecular patterns (PAMPS). PAMPs include cell membrane proteins, lipopolysaccharide and flagellular proteins. For these antigenic determinants immune cells carry pattern recognition receptors (PRRs) which bind non-specifically with corresponding PAMPs in order to generate immune response.

What is oxygen independent killing mechanism?

In this process of intracellular killing of bacteria, there is no need of oxygen. The granules and vesicles of phagocytes secrete various enzymes which are hydrolytic proteins .Those proteins are bacteriocidal in nature according to their modes of action.

Which defect are induced due to IgE deficiency?

IgE deficiency is a rare condition where the body doesn't produce enough Immunoglobulin E (IgE), a type of antibody that plays a key role in allergic reactions and parasite infections. Interestingly, IgE deficiency is not typically associated with increased susceptibility to infections or defects. In fact, people with IgE deficiency may have a lower risk of developing allergies and allergic diseases, such as asthma, atopic dermatitis, and food allergies.

Detail of digeorge syndrome:

DiGeorge Syndrome is a primary immunodeficiency disorder characterized by the complete absence of T-cells, also known as thymic aplasia or hypoplasia. It is linked with hypoparathyroidism and results from abnormal fetal development affecting the heart, thymus, and parathyroid glands. The condition is caused by a deletion on chromosome 22, which can vary in size but does not necessarily correlate with disease severity. Treatment often involves thymic grafts.

Adjuvant and its types?

These are the substances which increase the antigenicity of weak antigens. • Used with vaccine in vaccination • Alum or Aluminium hydroxide • Show undesirable side effects i.e fever

These can be of two forms 1) Chemicals 2) Biological.

How we can diagnose Type I hypersensitivity?

Type I hypersensitivity can be diagnosed using skin prick tests to identify allergen-specific reactions, serum IgE levels to detect allergen-specific antibodies, and patch tests to observe delayed responses. Additionally, blood tests such as the RAST (RadioAllergoSorbent Test) can help identify specific allergens.

Three accessory molecules?

The three accessory molecules are:

1. **CD4**: Binds to Class II MHC, aiding T-helper cell attachment to antigen-presenting cells.
2. **CD8**: Binds to Class I MHC, facilitating cytotoxic T-cell binding to target cells.
3. **CD2**: Interacts with Leucocyte Function Antigen 3 (LFA-3) on antigen-presenting cells.

.Antigenic shift and drift?

Antigenic shift This is the major form of change in antigenic structure of antigen as a result the new strains of microbes are generated.

Antigenic drift This is the minor form of antigenic variation within the same strain for example influenza (Flu) virus exhibits both antigenic shift and drift for pandemic and seasonal flu respectively. Similarly, bacteria like E.coli, Neisseria gonorrhoeae and Salmonella typhimurium also exhibit antigenic drift.

Components of CD3?

CD3 Composed of 5 proteins:CD3 complex

• One Gamma(γ) • One delta(δ) • Two Epsilon (ε) • Two Zeta (ζ)

Types of Freund adjuvants?

Freund's complete adjuvant (FCA): water-in-oil emulsion containing antigen with heat killed Mycobacterium tuberculosis • Stimulate both humoral & Cell mediated immune response for Ab • Used as booster antigen dose • Freund's adjuvant is used by mixing with equal parts of antigen.

Difference between primary and secondary antibody response?

The primary antibody response is the initial reaction to an antigen, characterized by the production of IgM antibodies, which have lower affinity and persistence. In contrast, the secondary antibody response involves a more robust reaction with IgG as the major antibody, alongside possible IgE or IgA, and shows higher affinity and persistence.

How cellular immunity protects the body:

Cellular immunity protects the body by activating T-cells to identify and destroy infected or abnormal cells. It involves cytotoxic T-cells killing infected cells, helper T-cells stimulating other immune responses, and regulatory T-cells maintaining immune system balance.

How parasite are killed mast cells and eosinophils.:

Mast cells and eosinophils collaboratively combat parasites. Mast cells release cytokines that activate eosinophils, enhancing their response. Eosinophils, in turn, release granules containing toxic proteins and enzymes that damage parasites. This mutual activation creates a hostile environment for the parasites, leading to their destruction. Mast cells and eosinophils thus work together to effectively neutralize and eliminate parasitic threats from the body.

Mechanism of antibody response of immune system.

The antibody response mechanism involves several key steps:

1. **Antigen Recognition**: B cells with specific receptors recognize and bind to antigens.
2. **Activation**: B cells are activated by antigen binding and helper T cells' signals.
3. **Proliferation**: Activated B cells proliferate and differentiate into plasma cells.
4. **Antibody Production**: Plasma cells secrete antibodies specific to the antigen.
5. **Antibody Action**: Antibodies neutralize pathogens, enhance phagocytosis, and activate the complement system.

Pathology defects of complement system?

• Acute Inflammatory response • Capillary dilatation • Exudation of plasma proteins & fluids (Edema) • Bronchoconstriction • Mast cells degranulation (Allergic reaction).

Difference isograft, allograft xenograft:

Isograft: Transplantation between genetically identical individuals, such as twins.

Allograft: Transplantation between individuals of the same species but with different genetic makeup.

Xenograft: Transplantation of cells, tissues, or organs across different species, such as from animals to humans.

Name the drug that induce hemolytic anemia?

The examples of exogenous antigens include Drug induced hemolytic anemia, granulocytopenia & thrombocytopenia.

Complement Fixation Test?

- **Test**: The complement fixation test is used to detect the presence of specific antibodies in a patient's serum. It measures the ability of these antibodies to fix complement proteins, leading to the lysis of target cells if the antibody is present.

Defects in IgA Deficiency?

- **Defects**: IgA deficiency can lead to recurrent infections, particularly in mucosal areas such as the respiratory and gastrointestinal tracts. Individuals may experience chronic sinusitis, bronchitis, and gastrointestinal infections due to the lack of IgA's protective function at mucosal surfaces.

Vaccine Used for Artificial Active Immunization?

- **Vaccine**: The MMR (measles, mumps, and rubella) vaccine is used for artificial active immunization, stimulating an immune response to protect against these diseases.

Examples of Endogenous Antigens?

- **Examples**: Endogenous antigens include normal cellular proteins and self-antigens that are expressed on the surface of cells, such as histones and other nuclear proteins.

MHC Class I and MHC Class II:

- **MHC Class I**: Present on all nucleated cells, presenting endogenous antigens to CD8+ cytotoxic T cells.
- **MHC Class II**: Present on antigen-presenting cells (APCs), presenting exogenous antigens to CD4+ helper T cells.

Disorder in MHC Class I:

- **Disorder**: Bare Lymphocyte Syndrome, where defective MHC Class I expression impairs cytotoxic T cell responses, leading to increased susceptibility to viral infections and some tumors.

Paracrine, Autocrine, Endocrine Definitions:

- **Paracrine**: Signaling where cells release substances affecting nearby cells.
- **Autocrine**: Signaling where cells release substances that affect themselves.
- **Endocrine**: Signaling where hormones are released into the bloodstream to affect distant organs.

Hybridoma Formation:

Hybridoma technology involves fusing antibody-producing B cells with myeloma cells to create hybrid cells (hybridomas) that produce specific monoclonal antibodies indefinitely.

Humoral Barrier:

Humoral barriers involve antibody-mediated protection by immunoglobulins (Ig) in bodily fluids, preventing pathogen entry and neutralizing toxins.

Diagnosis of Type I Hypersensitivity:

Diagnosed through skin prick tests or specific IgE blood tests to identify allergens causing immediate hypersensitivity reactions.

Anaphylaxis:

Anaphylaxis is a severe, rapid allergic reaction characterized by symptoms such as difficulty breathing, swelling, and hypotension. It requires immediate treatment with epinephrine.

Antigen Processing:

- **Endogenous Antigens**: Processed in the cytoplasm and presented by MHC Class I molecules to CD8+ cytotoxic T cells.
- **Exogenous Antigens**: Processed in endocytic compartments and presented by MHC Class II molecules to CD4+ helper T cells.

Role of Immune System in Transplant:

The immune system recognizes and rejects foreign tissues via cellular and humoral responses. Immunosuppressive drugs are used to prevent graft rejection and maintain transplant acceptance.